

Face Recognition

1. Experiment Objective

Use MediaPipe Face Detection to detect the positions of faces in the camera image, marking each face region with a rectangular bounding box and displaying the confidence.

2. Experiment Principle

1. Key Terms

Keyword	Description
MediaPipe Face Detection	Google's face detection model, which outputs a face bounding box + 6 keypoints (eyes, nose, mouth, ears) but not the 468-point mesh
Bounding box	The rectangular box enclosing the face region, in the format (x, y, width, height)
Confidence	The model's certainty about a detection result, overlaid on the frame as a percentage

2. Technical Architecture

```
Terminal 1: start agent + camera + joint models
Terminal 2: face recognition
```

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☒ Supported
No-PTZ version	☒ Supported
Required peripherals	ESP32 camera

2. Start the agent + camera + joint models (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```

yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-10-47-55-372666-yahboom-149737
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [149778]
[INFO] [camera_driver-2]: process started with pid [149780]
[INFO] [robot_state_publisher-3]: process started with pid [149782]
[INFO] [joint_state_publisher-4]: process started with pid [149784]
[INFO] [ekf_node-5]: process started with pid [149803]
[camera_driver-2] [INFO] [1785206876.886150022] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[joint_state_publisher-4] [INFO] [1785206877.007751174] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[micro_ros_agent-1] [1785206877.173080] info | UDPv4AgentLinux.cpp | init | running... | port: 8090
[robot_state_publisher-3] [INFO] [1785206877.754391885] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785206877.754890084] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785206877.754912957] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754918130] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785206877.754924428] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785206877.754929199] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785206877.754935071] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785206877.754939792] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785206877.754944035] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754949951] [robot_state_publisher]: got segment rfwheel_Link
[camera_driver-2] [WARN] [1785206879.871708745] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785206880.046238399] [esp32_ip_camera_publisher]: IP camera stream opened successfully

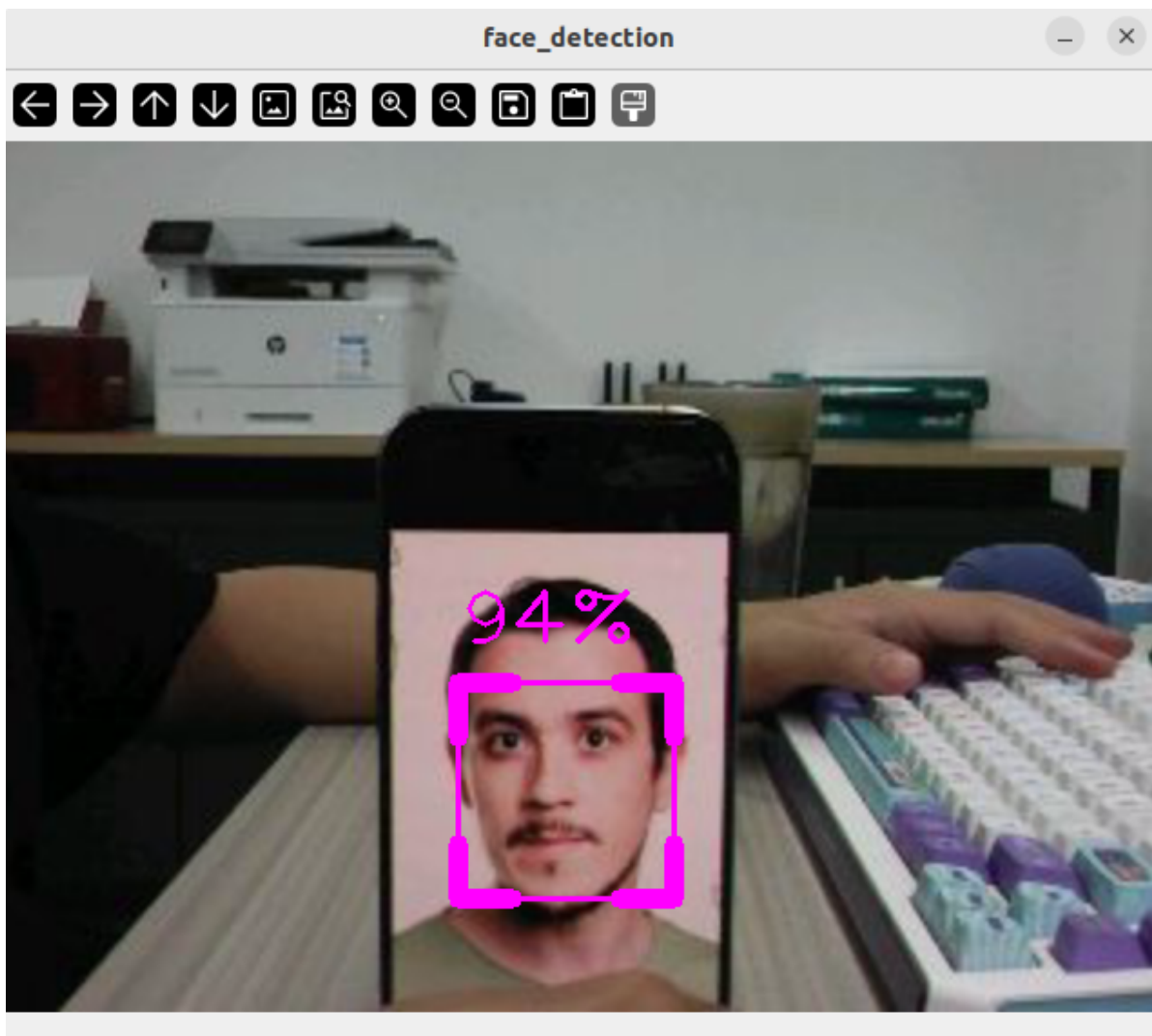
```

3. Start face recognition (Terminal 2)

```
ros2 launch microros_v2_mediapipe_pkg face_detection.launch.py
```

4. Observed Results

- Faces are enclosed in pink rectangular bounding boxes, with the confidence percentage shown above the box
- 6 keypoints (eyes, nose, mouth, ears) are annotated at their corresponding positions



5. Code Walkthrough

Source code paths:

- Launch:
`~/microros_ws/src/microros_v2_mediapipe_pkg/launch/face_detection.launch.py`
- Node:
`~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/FaceDetection.py`

1. Core Node Analysis

```
# Source file:
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/FaceDetection.py

class FaceDetection(Node):
    def __init__(self):
        super().__init__("face_detection_node")
        self.mp_face_detection = mp.solutions.face_detection
        self.face_detection = self.mp_face_detection.FaceDetection(
            model_selection=0, min_detection_confidence=0.5)

    def image_callback(self, msg):
        frame = self.bridge.imgmsg_to_cv2(msg, 'bgr8')
        results = self.face_detection.process(cv.cvtColor(frame,
            cv.COLOR_BGR2RGB))
        if results.detections:
            for detection in results.detections:
                bbox = detection.location_data.relative_bounding_box
                h, w, _ = frame.shape
                x, y = int(bbox.xmin * w), int(bbox.ymin * h)
                bw, bh = int(bbox.width * w), int(bbox.height * h)
                cv.rectangle(frame, (x, y), (x+bw, y+bh), (255, 0, 255), 2)
                cv.putText(frame, f'{detection.score[0]:.0%}',
                    (x, y-5), cv.FONT_HERSHEY_SIMPLEX, 0.5, (255, 0, 255), 2)
                cv.imshow(self.window_name, frame)
```

Key logic:

- `model_selection=0`: short-range model (within 2 meters), suitable for desktop/indoor scenes; `=1` is the long-range model
- `relative_bounding_box`: normalized coordinates (0~1) that must be multiplied by the actual image width/height to convert to pixel coordinates
- The bounding box + confidence percentage are drawn directly on the original frame; the 6 keypoints are obtained via `detection.location_data.relative_keypoints`

2. Key Parameters

Parameter definition files:

- Launch:
`~/microros_ws/src/microros_v2_mediapipe_pkg/launch/face_detection.launch.py`

- Node:

```
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/Face  
Detection.py
```

Parameter	Type	Default	Description
<code>model_selection</code>	int	0	0=short-range model, 1=long-range model
<code>min_detection_confidence</code>	float	0.5	Detection confidence threshold

3. Node Description

Node	Function
<code>face_detection_node</code>	MediaPipe Face Detection face bounding box + confidence annotation

6. Frequently Asked Questions

Issue	Cause	Solution
Face not detected at long range	<code>model_selection=0</code> only works within 2 meters	Set <code>model_selection:=1</code> to switch to the long-range model